THE UNIVERSITY OF MELBOURNE

COMP30023 – Computer Systems

# Internet (Network) Layer – IP Addresses and Packet Switching

Dr Lachlan Andrew

© 2025 University of Melbourne

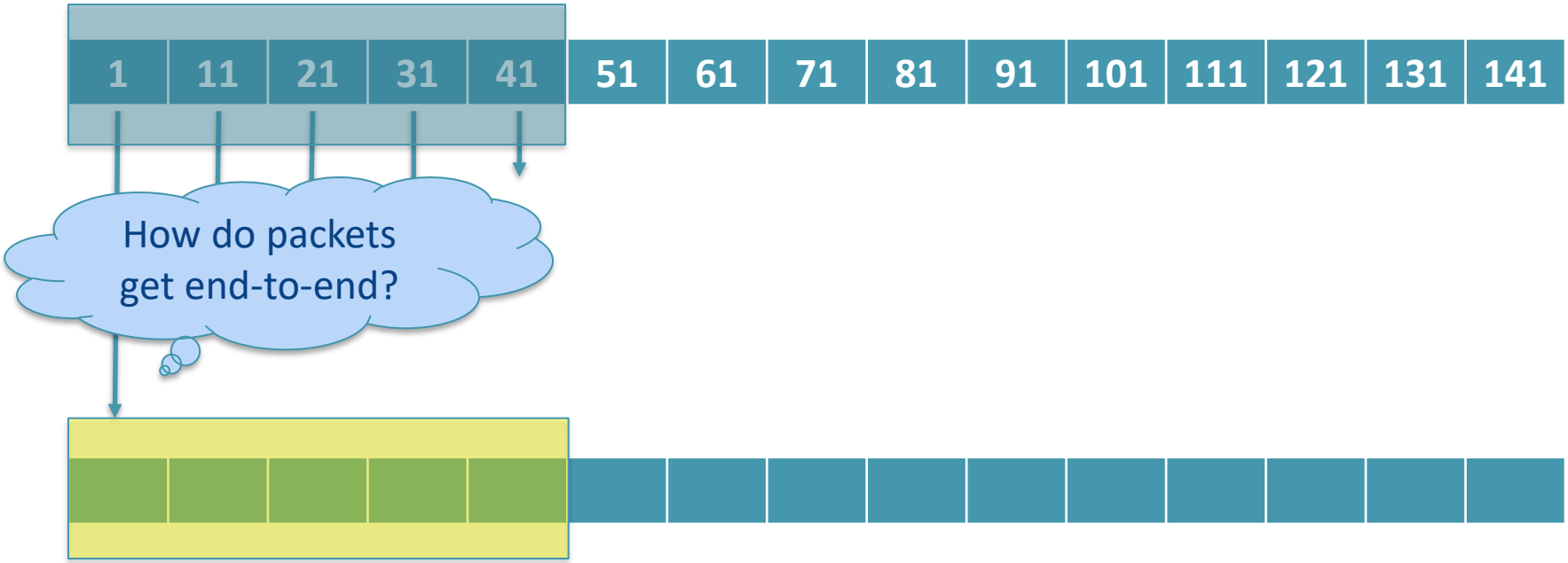
# Recap

Protocol design: Remote procedure call

- What RPC is
- Deciding services to offer
- Deciding packet format
- Deciding what information needs to be sent, when
- Deciding lower layer protocol to use

© 2025 University of Melbourne

# Recap: TCP Sliding Window




© 2025 University of Melbourne

# Summary

- Initial look at the network layer
- IP addressing

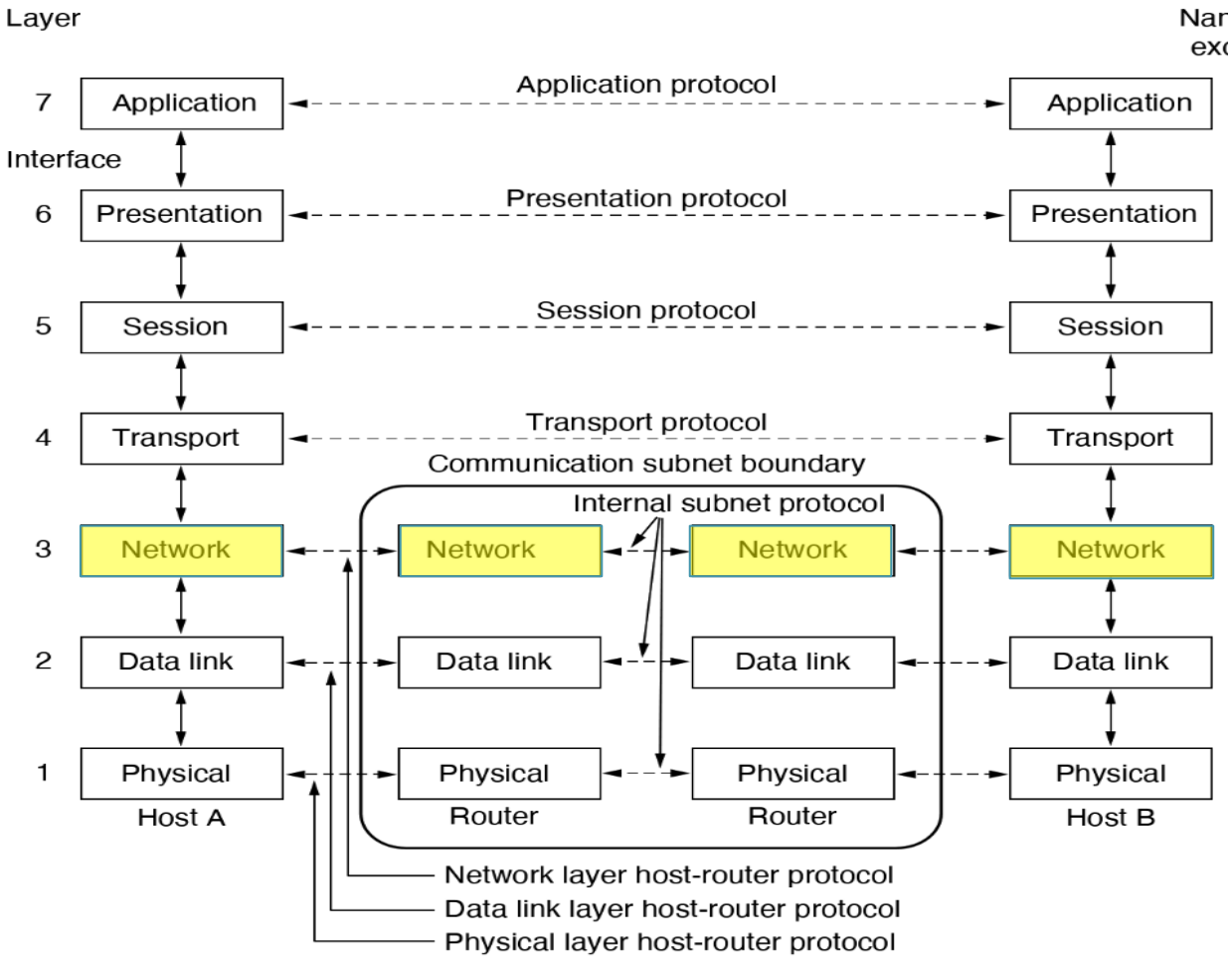
# Network addresses

```
    // #define required for VSCode
#define _POSIX_C_SOURCE 200112L
#include <netdb.h>
#include <string.h>
#include <stdio.h>
#include <unistd.h>

int connfd = 0, re = 1, s, n;
struct addrinfo hints, *res, *rp;

memset(&hints, 0, sizeof hints);
hints.ai_family = AF_INET6;
hints.ai_socktype = SOCK_STREAM;
hints.ai_flags = AI_ACTVE;
```

```
s = getaddrinfo("localhost",
                "5000", &hints, &res);
for (rp = res; rp != NULL;
            rp = rp->ai_next) {
    connfd = socket(rp->ai_family,
                    rp->ai_socktype,
                    rp->ai_protocol);
    if (connfd == -1) continue;
    if (connect(connfd,
                rp->ai_addr,
                rp->ai_addrlen) != -1
        break;
    close(connfd);
}
write (connfd, "Hello, network!\r\n");
close (connfd);
```

© 2025 University of Melbourne

# Position in the stack: Network layer



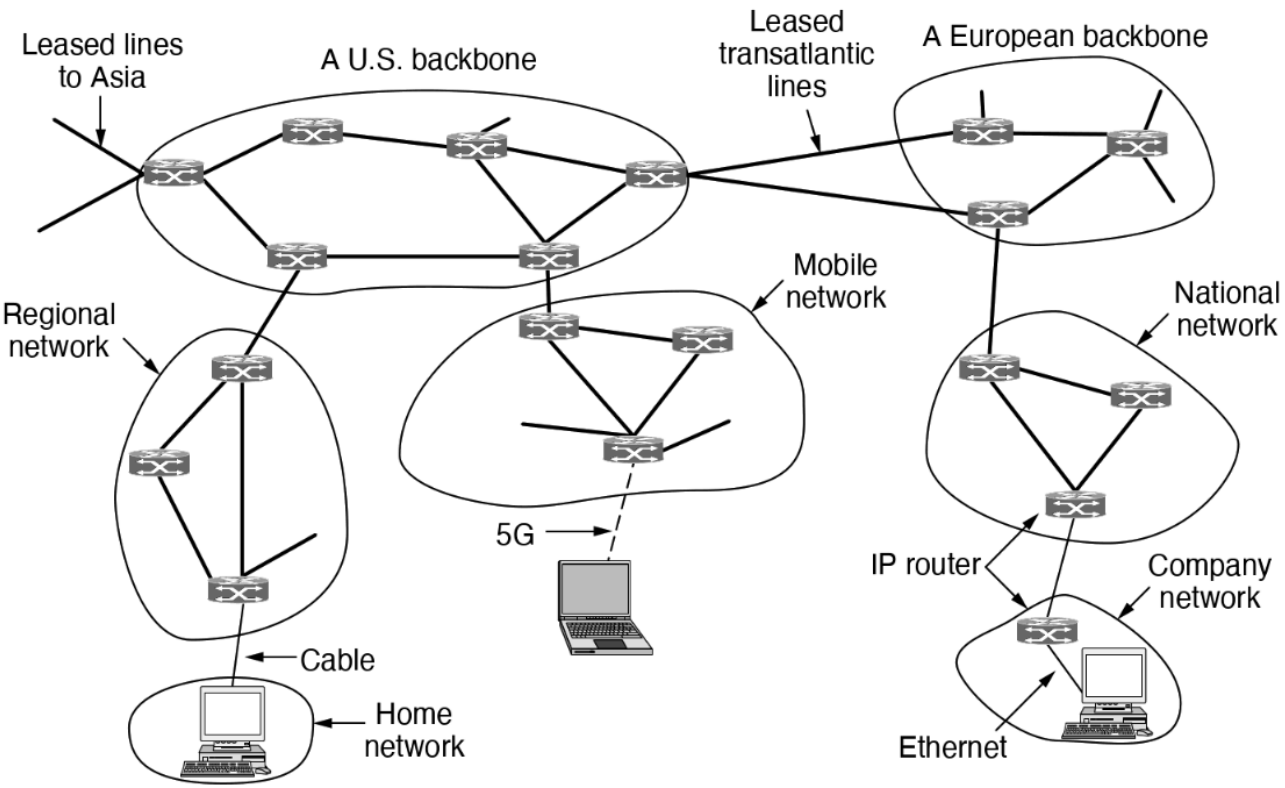

- Use data
- Tidy up e2e
- Get data e2e
- Tidy up p2p
- Get data p2p

# Internet (network) Layer

- Role: *get data from the source all the way to the destination*
  - May not be in a single hop (point-to-point link)
- Traffic must be *routed* efficiently
  - This is performed by network devices called routers
- Nodes must be given *names (addresses)*

# Internetworking and Routing

Connecting multiple networks



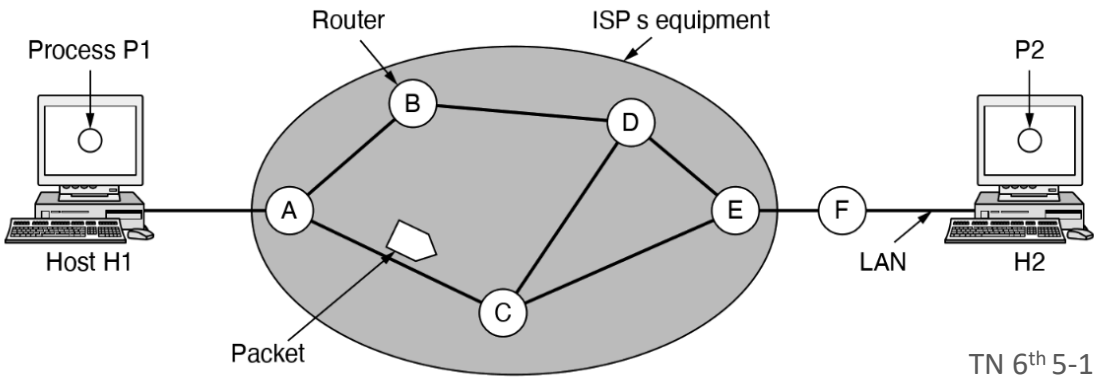

TN 6th 5-46

# Network Layer

- Most network layer code runs on routers
- Protocol data units are called “packets”
- What types of services does the network layer provide?
  - Connectionless
    - Packet switching (Internet Protocol - IP)
    - Minimum required service: “send packet”
    - Called “datagram” network
  - Connection-oriented
    - (virtual) Circuit Switching
      - Asynchronous Transfer Mode – ATM
      - MultiProtocol Label Switching MPLS
    - Called “virtual circuit” network
    - These usually act as a single “link” of an IP network

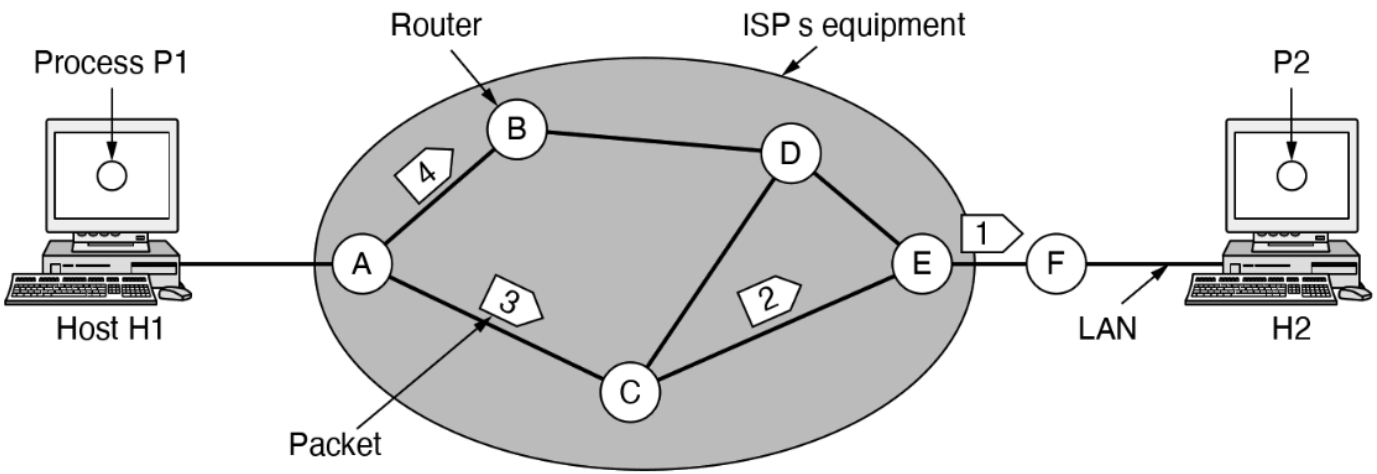
# Store-and-Forward Packet Switching

- The internet is a packet switched network
- Host H1 wants to send a packet to H2
  1. Transmits it to the nearest router (A)
  2. The packet is buffered while it is arriving, and the checksum is verified
  3. If valid, the packet is *stored* until the outgoing interface is free
  4. The router *forwards* the packet onto the next router in the path
  5. Repeat 2-4




TN 6th 5-1

# Packet forwarding – Connectionless (e.g., IP)




Paths can change for packets in the same transport layer connection

A s table (initially)

| Dest. | Line |
|---|---|
| A | – |
| B | B |
| C | C |
| D | B |
| E | C |
| F | C |

A s table (later)

| | |
|---|---|
| A | – |
| B | B |
| C | C |
| D | B |
| E | B |
| F | B |

C s table

| | |
|---|---|
| A | A |
| B | A |
| C | – |
| D | E |
| E | E |
| F | E |

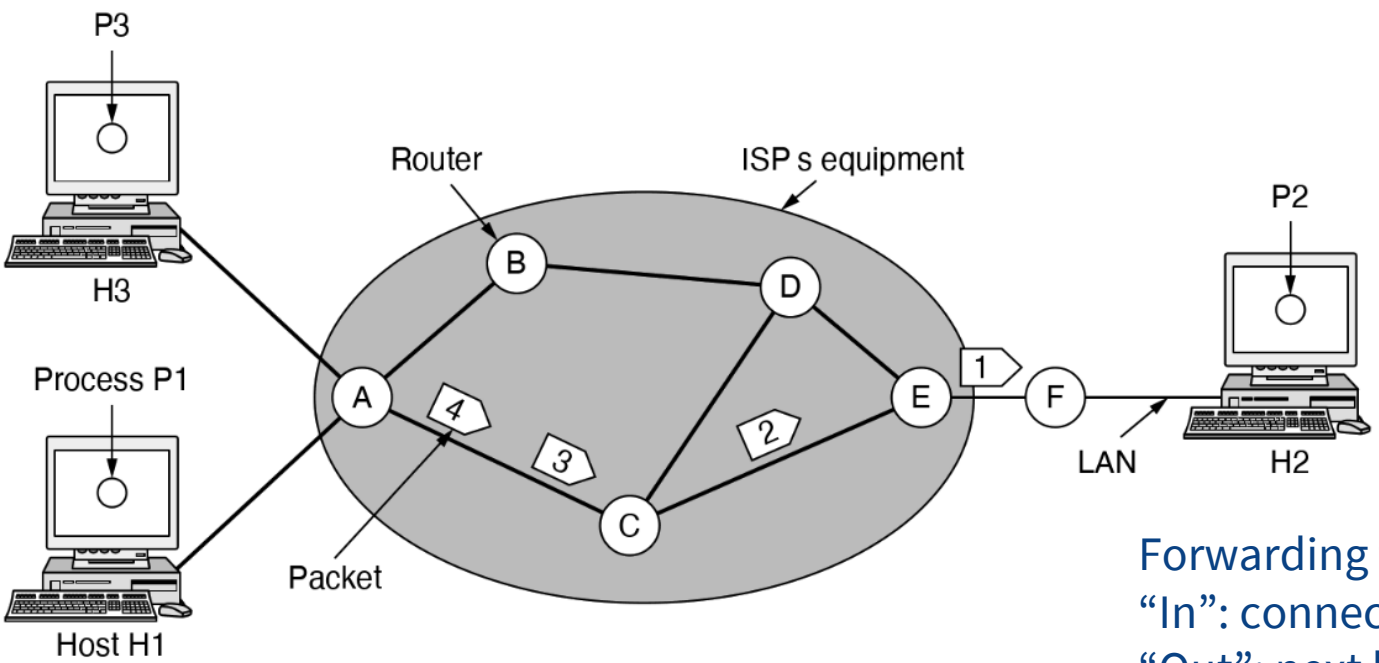
E s table

| | |
|---|---|
| A | C |
| B | D |
| C | C |
| D | D |
| E | – |
| F | F |

Forwarding tables (also called routing tables)

TN 6th 5-2

# Packet forwarding – Connection-oriented




A s table

| In | | Out | |
|---|---|---|---|
| H1 | 1 | C | 1 |
| H3 | 1 | C | 2 |

C s table

| | | | |
|---|---|---|---|
| A | 1 | E | 1 |
| A | 2 | E | 2 |

E s table

| | | | |
|---|---|---|---|
| C | 1 | F | 1 |
| C | 2 | F | 2 |

Virtual circuit number

Forwarding table:
"In": connection ID
"Out": next hop and new connection number

VC number is local to a hop

TN 6th 5-3

# Pros and cons of connections

| Issue | Datagram Network | Virtual Circuit |
|---|---|---|
| Type | Connectionless | Connection-oriented |
| State | + Routers do not hold state information about connections | - Each VC requires router table space.<br>- Router reboots a problem. |
| Addressing | (-) Each packet has full source and destination | (+) Each packet contains a short VC number |
| Routing | Each packet independently | Defined at set-up |
| Quality of Service | - Difficult | + Easy if enough resources |
| Congestion control | - Difficult | + Easy if enough resources |
| Link failure recover | + Simple | - Extra work |

# Internet Protocol

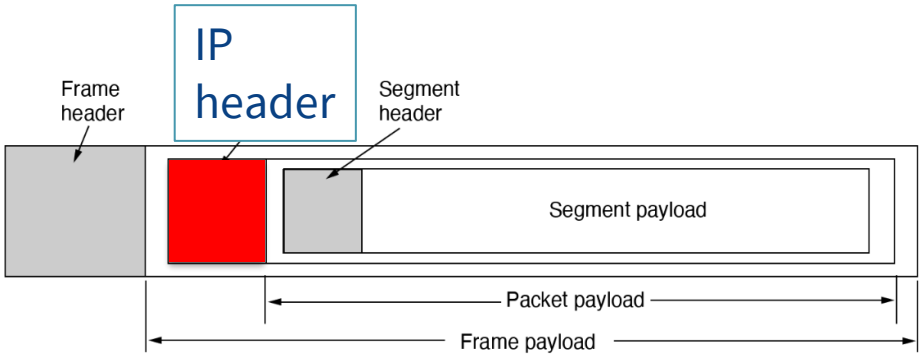
- Design principles:
  - Something that works OK is better than an ideal standard “in progress”
    - Standard of “OK” has gone up as the Internet has become vital
  - Keep it simple – Occam’s Razor
  - Be strict when sending and tolerant when receiving
    - E.g., web browsers handle pages with invalid HTML
  - Make clear choices – don’t have different approaches in a standard
  - Avoid static options and parameters – negotiate them at runtime
  - (New principle: Think about scalability)
- “Best effort”, not guaranteed performance

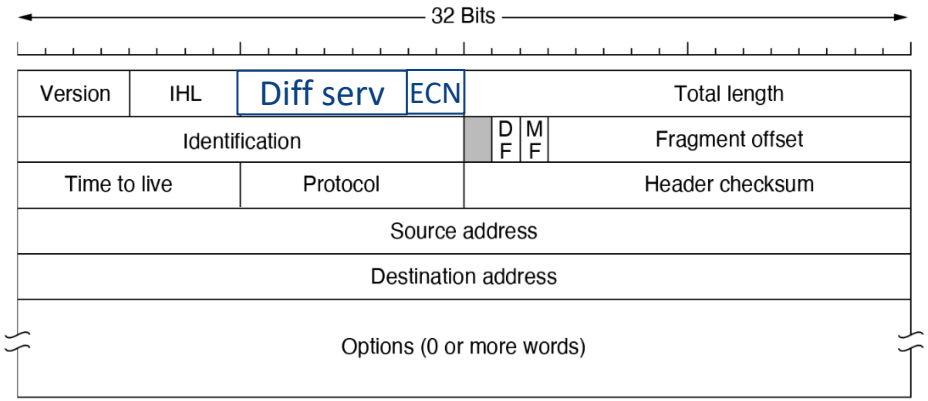
# Internet Protocol

- Responsible for moving the packets through the various networks from source to destination host
- Multiple paths through the network
  - Important for redundancy
- Routing algorithms are used to determine best path
  - We will look into these more later

# IP Version 4 Protocol (IPv4)

| Field | Usage |
|---|---|
| Version | Protocol version 4<br>(This field is also in IPv6) |
| IHL | Header length in 32 bit words; min 5, max 15 |
| Differentiated services | QoS service class |
| ECN | Explicit congestion notification |
| Total length | Including payload, max 65,535 |
| Identification, DF, MF, Fragment Offset | Used in the handling of fragmentation |
| Time to live (TTL) | Countdown of hops, at zero packet is discarded |
| Protocol | Transport layer service (TCP/UDP/SCTP/DCCP/etc.) |
| Source and Destination | IPv4 address |
| Options | Rarely used and poorly supported |




TN 6th 5-47

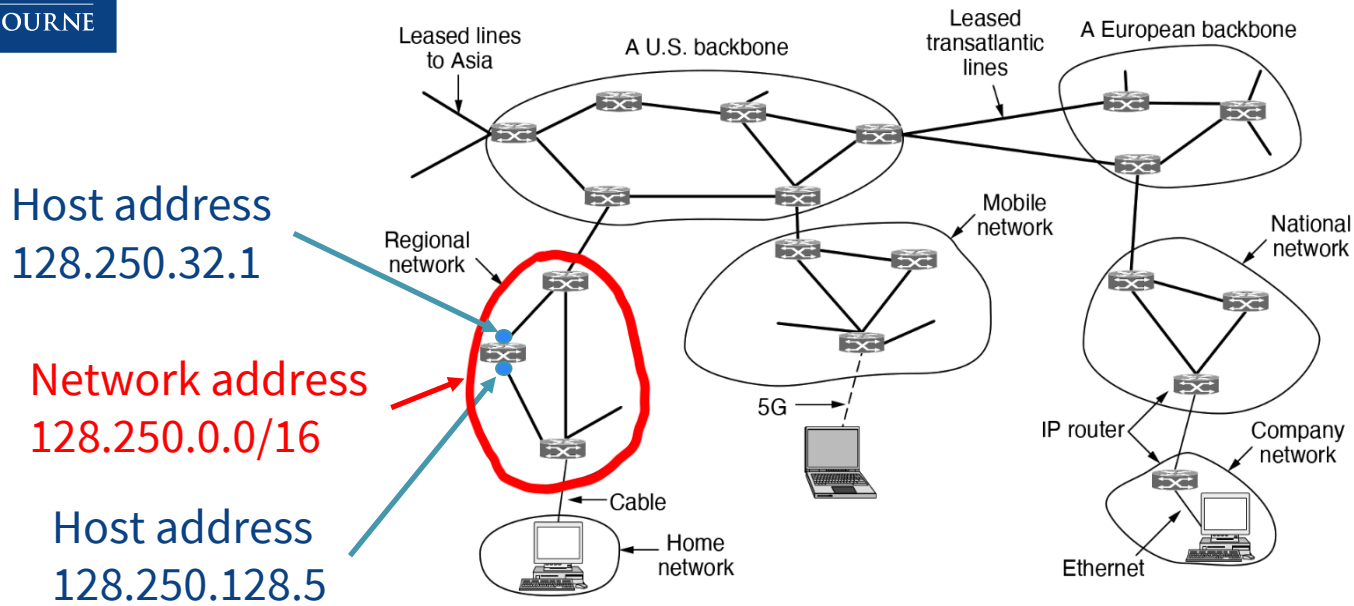
# IPv4 Addresses

- 32-bit number
- Expressed in decimal notation, each byte is shown as a decimal, separated by a period, 172.22.44.10 (0xAC162C0A)
- 0.0.0.0 is lowest, 255.255.255.255 is highest
- IP addresses are given to *interfaces* not hosts, i.e. a host with multiple network cards will have multiple IP addresses
- Supply of IPv4 addresses has basically been exhausted

# Types of address

- Unicast:  One destination  (“normal” address)
- Broadcast: Send to everyone
- Multicast: Send to a particular set of nodes
  - Used for streaming video of live events

# Network and Host Addresses




- IP address bits split into “network address” + “host address”
- Scalable
  - Send a packet to the right network
  - Let the network get it to the right host

# Network and Host Addresses

- Hierarchical - encodes the network and host number
  - Network in top bits
  - Host in bottom bits

10000000.11110101.11000000.00000001

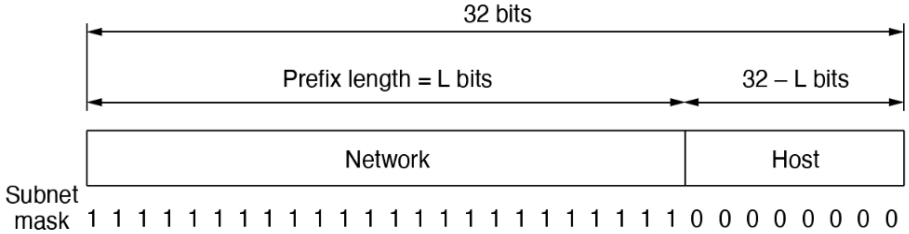
10000000.11110101.11000000.00000001

- Assigned to networks in blocks, *the network part will be the same for all hosts on that network*
  - A network corresponds to a contiguous block of IP address space, called a prefix
  - Prefixes are written as the lowest IP address followed by a slash and the size of the network portion - 192.0.2.0/24
  - An interface must “know” the size of its network part

# IP Addresses - Blocks

Smaller number
=
Bigger network

- In the case of 192.0.2.0/24
  - 24 bits are for the network 192.0.2.0
  - Leaving 8 bits for hosts – up to 256 addresses
- In the case of 10.0.0.0/8 (reserved private block)
  - 8 bits are for the network 10.0.0.0
  - Leaving 24 bits for hosts – up to 16,777,216 addresses
- Can also be written as a subnet mask, a binary mask of 1's
  - In the case of /24: the subnet mask is 255.255.255.0




TN 6th 5-49

# Exercises

- Is 128.250.73.5 in 128.250.0.0/16?
- Is 128.250.73.5 in 128.250.0.0/24?
- Is 128.250.73.5 in 128.250.0.0/17?
- How many addresses are in 128.250.0.0/17?
- What is the network mask of 128.250.73.5?

# Interfaces must know netmask

- The last exercise was a trick question
- We can’t tell the network mask from the IP address
- Each interface needs to be told both
  - IP address  AND
  - Network mask (must be the same as all others on its network)

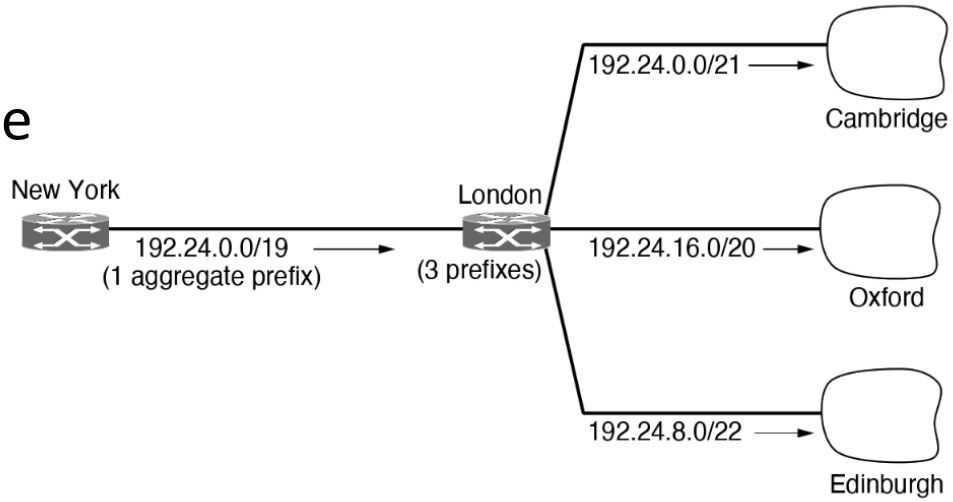
# IP Addresses – Prefixing

- Network number = network mask (bitwise-AND) IP address

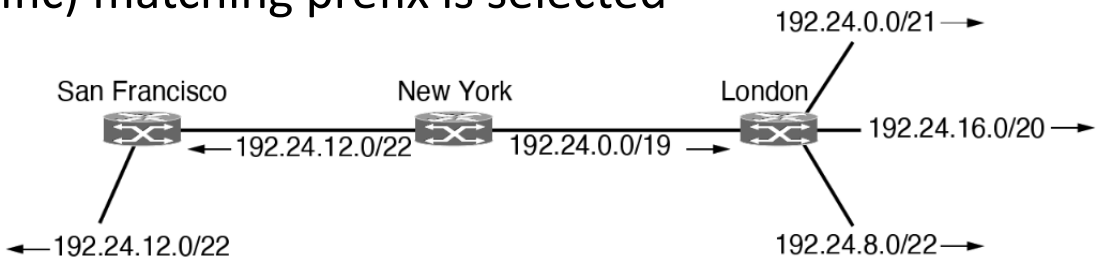
- This is crucial for efficient routing on the internet
  - Networks are assigned blocks of addresses → intermediate routers need only maintain routes for the prefixes, not every individual host
  - Only when the packet arrives at the destination network does the host portion need to be read

# IP Addresses – Route aggregation

- Aggregation is performed automatically
- Currently it roughly halves the size of the routing table

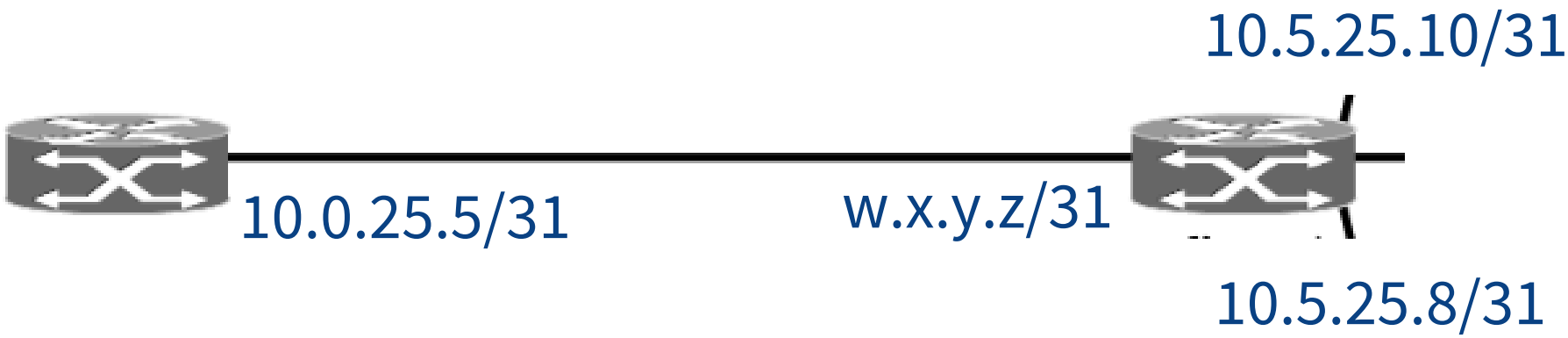



- Prefixes can overlap
  - Longest (most specific) matching prefix is selected




TN 6th 5-52, 5-53

# Routers *join* networks



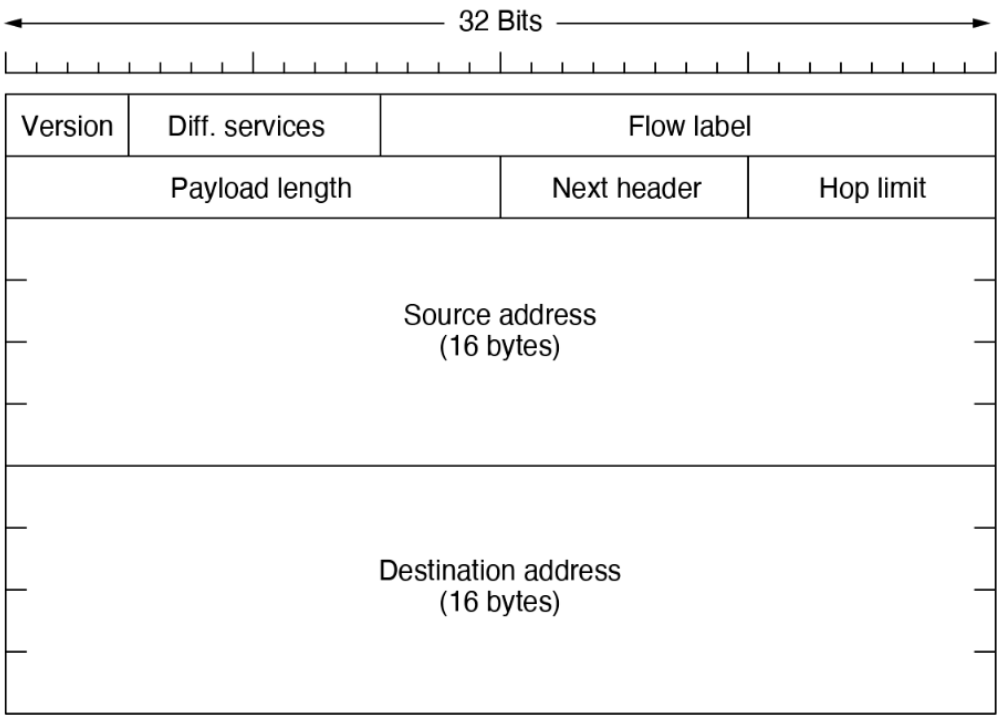

- Routers are special nodes that join networks
  - A link between routers is a “network” all by itself
- Above, work out what  w, x, y and z are
  - Hint: What are the network bits?
- Notice that the router has many IP addresses.  Why?

# IPv6

- Designed over 20 years ago to address the problem of exhausting the IPv4 address space
- While solving that problem, some other changes were made
  - Simpler header – allows faster processing
  - Improved security – now back-ported to IPv4
  - Further Quality of Service support
- IPv6 addresses are 128 bits
  - Unlikely ever to run out

# IPv6 Header

| Field | Usage |
|---|---|
| Version | 6 |
| Differentiated services | 6 bits for service class, 2 bits for congestion control (ECN) |
| Flow label | Pseudo-Virtual Circuit identifier |
| Payload length | Bytes after the 40 byte header |
| Next header | Used to specify additional headers or Protocol (TCP/UDP) |
| Hop limit | Same as TTL (Time To Live) |
| Source and Destination | 16 byte IPv6 addresses |




TN 6th 5-57

# IPv6 Addressing

- Written as 8 groups of (up to) 4 hex digits
  - 8000:0000:0000:0000:0123:4567:89AB:CDEF
  - [8000:0000:0000:0000:0123:4567:89AB:CDEF]:port_number
- Can be optimised by stripping *one* group of consecutive 0's
  - 8000::123:4567:89AB:CDEF
- Backwards compatibility with IPv4 is achieved with
  - ::ffff:192.31.2.46 (note the mix of hex with decimal)
- Still not widely deployed
  - Google measures IPv6-*capable* requesters, https://www.google.com/intl/en/ipv6/statistics.html
  - still around 40% globally, (~32% in Australia)
- Fairly widely supported – likely to see a continued growth

# IPv6 addresses are larger than IPv4

```
    // #define required for VSCode
#define _POSIX_C_SOURCE 200112L
#include <netdb.h>
#include <string.h>
#include <stdio.h>
#include <unistd.h>

int connfd = 0, re = 1, s, n;
struct addrinfo hints, *res, *rp;

memset(&hints, 0, sizeof hints);
hints.ai_family = AF_INET6;
hints.ai_socktype = SOCK_STREAM;
hints.ai_flags = AI_ACTVE;
```

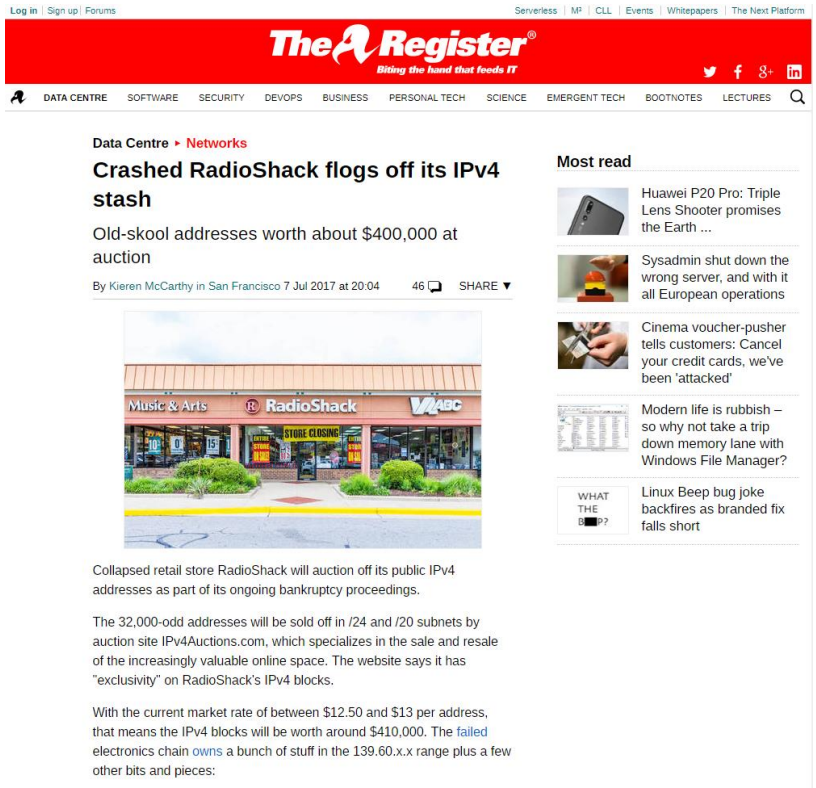
```
s = getaddrinfo("localhost",
                "5000", &hints, &res);
for (rp = res; rp != NULL;
            rp = rp->ai_next) {
    connfd = socket(rp->ai_family,
                    rp->ai_socktype,
                    rp->ai_protocol);
    if (connfd == -1) continue;
    if (connect(connfd,
                rp->ai_addr,
                rp->ai_addrlen) != -1
        break;
    close(connfd);
}
write (connfd, "Hello, network!\r\n");
close (connfd);
```

Some, like phone, have variable sizes

© 2025 University of Melbourne

# And finally…

- One of the disadvantages of a hierarchical address space is that it can be quite wasteful if not carefully assigned, i.e. large parts of the address space unused if blocks badly assigned
- Combined with the scarcity (exhaustion) of available IP addresses, they have become a valuable commodity
- Not intended to be sold, should really be returned to the assigning body for reallocation
- Early adopters of IPv6 were able to sell their IPv4 address space at a premium



Log in | Sign up | Forums

Serverless | MP | CLL | Events | Whitepapers | The Next Platform

The Register®
Biting the hand that feeds IT

DATA CENTRE SOFTWARE SECURITY DEVOPS BUSINESS PERSONAL TECH SCIENCE EMERGENT TECH BOOTNOTES LECTURES

Data Centre ▸ Networks

## Crashed RadioShack flogs off its IPv4 stash

Old-skool addresses worth about $400,000 at auction

By Kieren McCarthy in San Francisco 7 Jul 2017 at 20:04 46 SHARE ▼



Collapsed retail store RadioShack will auction off its public IPv4 addresses as part of its ongoing bankruptcy proceedings.

The 32,000-odd addresses will be sold off in /24 and /20 subnets by auction site IPv4Auctions.com, which specializes in the sale and resale of the increasingly valuable online space. The website says it has "exclusivity" on RadioShack's IPv4 blocks.

With the current market rate of between $12.50 and $13 per address, that means the IPv4 blocks will be worth around $410,000. The failed electronics chain owns a bunch of stuff in the 139.60.x.x range plus a few other bits and pieces:

Most read

Huawei P20 Pro: Triple Lens Shooter promises the Earth ...

Sysadmin shut down the wrong server, and with it all European operations

Cinema voucher-pusher tells customers: Cancel your credit cards, we've been 'attacked'

Modern life is rubbish – so why not take a trip down memory lane with Windows File Manager?

Linux Beep bug joke backfires as branded fix falls short

# Acknowledgement


- The slides are based on slides prepared by Chris Culnane based on material developed previously by: Michael Kirley, Zoltan Somogyi, Rao Kotagiri, James Bailey and Chris Leckie.
- Some of the images included in the notes were supplied as part of the teaching resources accompanying the text books listed in lecture 1.
  - (And also) Computer Networks, 6th Edition, Tanenbaum A., Wetherall. D. https://ebookcentral.proquest.com/lib/unimelb/detail.action?docID=6481879
- Textbook Reference: 4.1, 4.2.1, 4.3.1, 4.3.3, 4.3.5